

5-Mark Questions & Answers — DBMS Module 5: Query Optimization & Transaction Concurrency Control

Q1. Explain cost estimation for selection and join operations in query optimization.

1. **Selection Cost** (b_r = blocks of r):
 - Equality condition on non-index: Cost = b_r block accesses.
 - Secondary Index lookup: Cost = Index Depth + 1.
2. **Join Cost** (r rowtiefen):
 - **Nested-Loop Join**: Cost = $b_r + (n_r \times b_s)$ block reads.
 - **Block Nested-Loop Join**: Cost = $b_r + (b_r \times b_s)$ block reads.
 - **Hash / Merge Join**: Cost = $3(b_r + b_s)$ block accesses.

Q2. Explain 2-Phase Locking Protocol (2PL) and Multiple Granularity Locking (IS, IX, SIX).

- **2PL**: Growing Phase (Acquire locks) -> Lock Point -> Shrinking Phase (Release locks). Guarantees conflict serializability.
- **Multiple Granularity Locking**: Hierarchy (Database -> File -> Page -> Record). Uses Intent locks:
- **IS**: Intent to acquire Shared locks on children.
- **IX**: Intent to acquire Exclusive locks on children.
- **SIX**: Shared lock on current subtree + IX on child nodes.

Q3. Explain Timestamp Ordering Protocol and Thomas Write Rule.

- **Timestamp Ordering**: Assigns $TS(T_i)$. Checks $TS(T_i) < W-TS(X)$ for Read and $TS(T_i) < R-TS(X)$ for Write. Aborts if violation occurs (Deadlock Free).
- **Thomas Write Rule**: If $TS(T_i) < W-TS(X)$ during $Write(X)$, the write request is obsolete and simply **ignored** rather than aborting T_i .

Q4. Explain Log-Based Recovery (Deferred vs Immediate update) and Checkpointing.

- **Deferred Update (NO-UNDO/REDO)**: Writes postponed until Commit. Requires only REDO during crash recovery.
- **Immediate Update (UNDO/REDO)**: Writes allowed while active. Requires UNDO for active and REDO for committed transactions.
- **Checkpointing**: Periodically flushes dirty buffers to disk, limiting crash recovery scan depth.

Q5. Describe Two-Phase Commit (2PC) protocol for distributed transactions and TCL commands.

- **2PC Protocol**:
 1. **Phase 1 (Prepare)**: Coordinator sends **PREPARE**; participants vote **VOTE_COMMIT** or **VOTE_ABORT**.
 2. **Phase 2 (Commit)**: If all voted commit, coordinator sends **GLOBAL_COMMIT**; else **GLOBAL_ABORT**.
- **TCL Commands**: **COMMIT** (Save), **ROLLBACK** (Undo), **SAVEPOINT** (Checkpoint inside transaction).

